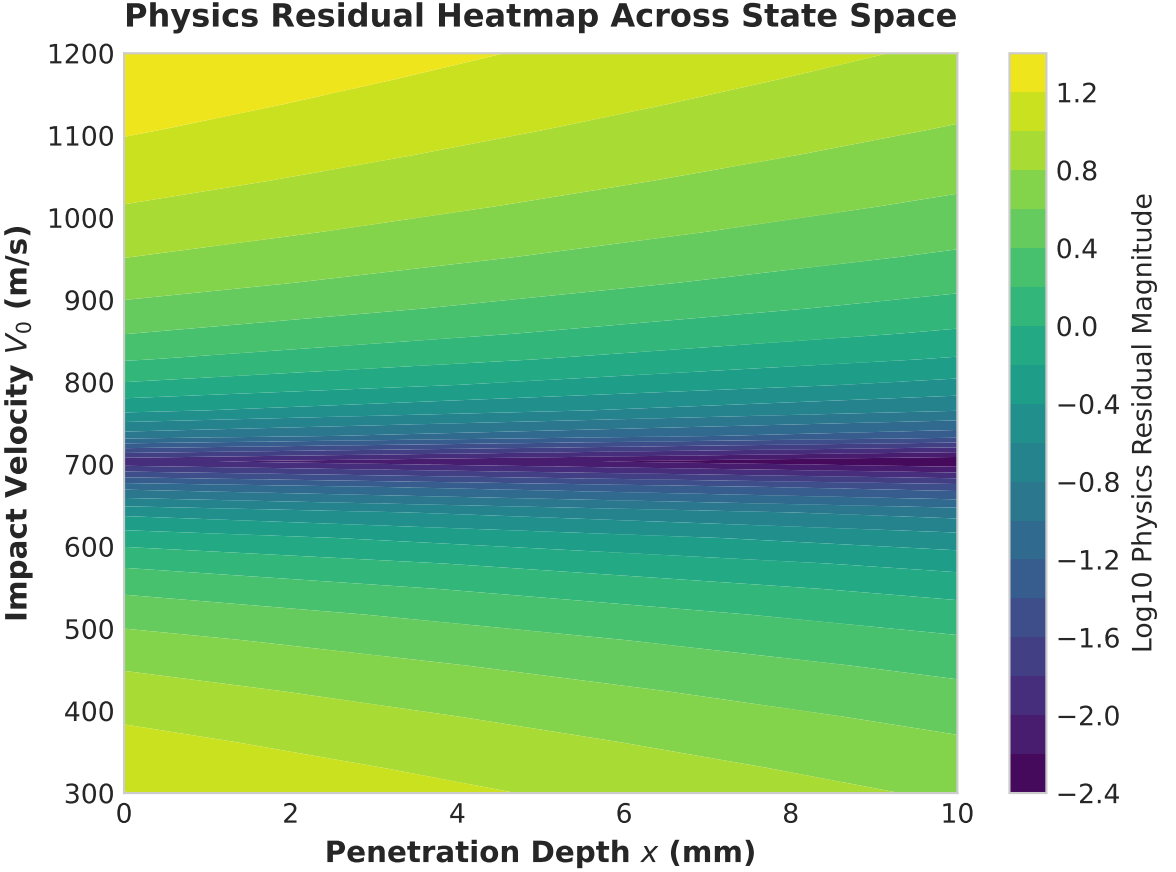
Physics Residual Heatmap Across State Space
Impact Velocity V_0 (m/s)
1200
1100
1000
900
800
700
600
500
400
300
0
2
4
6
8
10
Penetration Depth x (mm)
Log10 Physics Residual Magnitude
1.2
0.8
0.4
0.0
−0.4
−0.8
−1.2
−1.6
−2.0
−2.4